

Coin Rise in Viscous Fluid Worksheet

Measuring Terminal Velocity and Fluid Viscosity

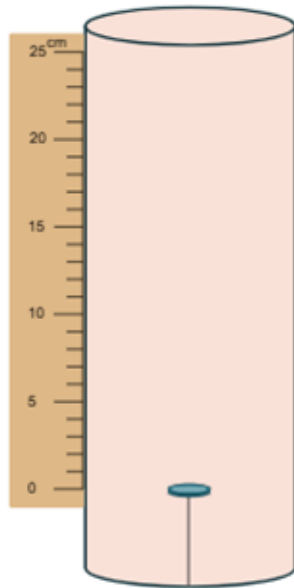
Name(s): _____

Learning Objectives

1. Define terminal velocity and describe how to measure it experimentally.
2. Compare the terminal velocity of a disk falling face-down versus edge-on.
3. Use the measured terminal velocity to estimate the viscosity of the fluid.

Experimental Setup

A 3D-printed polystyrene quarter is released into a column of viscous corn syrup, either **face-down** (disk face perpendicular to gravity) or **edge-on** (disk face parallel to gravity). Because the coin is less dense than the syrup, it rises until it reaches a terminal velocity, which is determined by recording distance traveled versus elapsed time. The terminal velocity is used to infer the fluid viscosity.



Before the Experiment

1. Which orientation do you predict will reach a higher terminal velocity: face-down or edge-on? Explain your reasoning in terms of drag forces.

2. The fluid in the column is corn syrup, which is denser than polystyrene. In which direction (up or down) will the coin move? What would happen if the same experiment were performed in water instead? Would the direction and terminal speed be the same? Explain.

3. Either viscous forces ($F_{drag} \propto \mu U$) or inertial forces ($F_{drag} \propto \rho U^2$) can dominate the drag force. These regimes are distinguished by the Reynolds number $Re = \rho a U / \mu$. How will you determine which regime applies here?

Table 1. Key parameters used in the analysis

Symbol	Description	Value	Unit
a	Radius of coin (look up)		cm
w	Thickness of coin (look up)		cm
ρ_e	Density of coin (look up)		g cm^{-3}
ρ	Density of corn syrup	1.33	g cm^{-3}
μ	Viscosity of fluid (to find)	—	cP
g	Gravitational constant	980	cm s^{-2}

Notation

d = distance coin traveled (cm)

t = time (s)

U_{fd} = terminal velocity for coin face-down (cm/s)

U_{eo} = terminal velocity for coin edge-on (cm/s)

F_{fd} = Stokes-flow drag force for coin face-down ()

F_{eo} = Stokes-flow drag force for coin edge on ()

Running the Experiment

1. Set the coin orientation to face-down, press the green Release button and start a stopwatch. Measure the elapsed time and distance for 6 or 7 times. Record the values in Table 2.

Table 2. Face-down orientation

[illegible]

2. Set the coin orientation to edge-down, press the green Release button and start a stopwatch. Measure the elapsed time and distance for 6 or 7 distances. Record the values in Table 3.

Table 3. Edge-On Orientation

[illegible]

After the Experiment

1. **Velocity from data.** Plot distance versus time for each orientation (or compute successive slopes $\Delta d/\Delta t$). At what time does each curve become approximately linear? Estimate the terminal velocity U_{fd} (face-down) and U_{eo} (edge-on) from the late-time slope.

Face-down: $U_{fd} \approx$ _____ cm s⁻¹ Edge-on: $U_{eo} \approx$ _____ cm s⁻¹

Is one faster than the other? Does this match your prediction from Question 1?

2. **Force balance and theoretical terminal velocities.** At terminal velocity, the net buoyant force equals the drag. The Stokes-flow drag on a thin disk of radius a is:

$$F_{fd} = 16\mu a U_{fd} \qquad F_{eo} = (32/3)\mu a U_{eo}$$

Perform a force balance to derive expressions for U_{fd} and U_{eo} in terms of ρ_e, ρ, a, w, μ , and g . Show your work.

What is the predicted ratio U_{fd}/U_{eo} ? Does your measured ratio agree?

3. **Coin dimensions.** Look up the radius a and thickness w of a US quarter and density ρ_e of polystyrene. Record the values in Table 1.

4. **Viscosity.** Use your measured terminal velocity and the expression from Question 2 to solve for μ . Compute μ from *both* orientations.

From face-down: $\mu \approx$ _____ cP From edge-on: $\mu \approx$ _____ cP

5. **Reynolds number check.** Calculate the Reynolds number $Re = \rho a U / \mu$ for each orientation. Is $Re \ll 1$? Does this justify the Stokes drag formulae in Question 2?

6. List reasons why the viscosity calculated from the face-down and edge-on orientations may not agree exactly.

7. Would the coin move faster or slower if the corn syrup were warmer? Explain.

8. If this experiment was carried out in water instead of corn syrup, what would happen. Why?